fMRI Foundation Model (V2) — Results

In every table, green marks the best value of the row (in the SOTA tables: the per-metric winner between us and the paper).

1. Pretraining corpus

Five sources, **4627 scans**, harmonized to TR = 0.72 s with a fixed T = 270 window.

Dataset	Native TR (s)	Probe holdout (30%)	Per-batch quota
HCP	0.72	yes	4
ABIDE	per-site	yes	4
OASIS-3	2.2	yes	4
ADNI	3.0	yes	3
AOMIC	0.75/2.0	no (kept whole)	1

Quota = per-batch proportion. Holdout: 30% of subjects excluded from pretraining -> leakage-free test set.

2. Datasets & scan counts

Dataset	Role	Scans	Note
HCP (rest)	pretrain + probe	1084	Sex / Age
ABIDE	pretrain + probe	1035	Autism / Age / Sex
OASIS-3	pretrain + probe	1197	AD Conversion (labels pending)
ADNI	pretrain + probe	812	NC-MCI / AD-HC / Amyloid
AOMIC	pretrain only	~499	derived (4627 - others)
ADHD-200	probe only (external)	162	115 with usable DX
COBRE	probe only (external)	146	schizophrenia (72 SZ / 74 HC)
UCLA (ds000030)	probe only (external)	265	downloader ready
HCP task-fMRI	probe only	—	7 tasks, download pending

External datasets (ADHD-200 / COBRE / UCLA) were never seen in pretraining.

3. Pretraining ablations (5 SSL runs) — test AUROC

Same DINOv2 (ImageNet) init; each run changes **one** factor. **base** = reference, **fourier** = Fourier positional encoding, **noblock2** = drop block_2, **pool** = AvgPool downsampling, **unfrozen** = all layers unfrozen during SSL. Green = best run for that axis.

Axis	**base**	**fourier**	**noblock2**	**pool**	**unfrozen**
ABIDE / Autism	0.605	0.518	0.484	0.606	0.541
ABIDE / Age	0.846	0.756	0.776	0.877	0.828
ABIDE / Sex	0.521	0.514	0.492	0.694	0.538
ADNI / NC-MCI	0.567	0.497	0.484	0.566	0.604
ADNI / AD-HC	0.481	0.537	0.577	0.591	0.731
ADNI / Amyloid	0.611	0.539	0.500	0.641	0.640
HCP / Sex	0.908	0.900	0.933	0.962	0.886
HCP / Age	0.633	0.594	0.626	0.680	0.592
OASIS / AD Conv	–	–	–	–	–

AUROC ranks the runs by representation quality (threshold- and balance-independent). The SOTA table uses Acc/F1, matching the papers.

4. Probe ablations (on `base`)

4a. Temporal aggregation of the CLS token — AUROC

Axis	mean (384-d)	mean_std (768-d)	Δ
ABIDE / Autism	0.605	0.583	-0.023
ABIDE / Age	0.846	0.835	-0.011
ADNI / NC-MCI	0.567	0.505	-0.062
ADNI / Amyloid	0.611	0.606	-0.005
ADNI / AD-HC	0.481	0.404	-0.077
HCP / Sex	0.908	0.853	-0.055
HCP / Age	0.633	0.595	-0.037
ADHD / ADHD	0.552	0.557	+0.005
COBRE / Schizophrenia	0.519	0.538	+0.019

`mean_std` mainly helps task dynamics; elsewhere `mean` wins.

4b. MLP head architecture — AUROC

Axis	128	256	256×128	512×256	512×256×128	linear
ADNI / NC-MCI	0.588	0.509	0.523	0.475	0.433	0.567
ADNI / AD-HC	0.407	0.467	0.472	0.448	0.487	0.481
ADNI / Amyloid	0.621	0.623	0.610	0.631	0.610	0.611
ABIDE / Autism	0.550	0.497	0.561	0.509	0.522	0.605
HCP / Sex	0.844	0.906	0.896	0.898	0.895	0.908
ADHD / ADHD	0.477	0.466	0.382	0.412	0.397	0.552
COBRE / Schizophrenia	0.602	0.566	0.595	0.557	0.590	0.519

The MLP head does not clearly beat the linear probe; deeper heads overfit.

5. Our results vs SOTA (same-dataset only)

5a. Same-dataset comparisons

SOTA shown only where the paper uses our dataset. Brain-JEPA = fine-tuning, Acc/F1 only; ours = linear probe (best-AUROC config). Green = per-metric winner.

Benchmark	Model	Acc	F1	AUROC
ADNI / NC-MCI	Ours (lin.)	0.576	0.632	0.604
	Brain-JEPA (FT)	0.768	0.863	n/r
ADNI / Amyloid	Ours (lin.)	0.559	0.623	0.641
	Brain-JEPA (FT)	0.710	0.760	n/r
ADHD-200	Ours (lin.)	0.617	0.287	0.557
	NeuroSTORM	0.587	n/r	n/r

5b. Our other downstream results (no same-dataset SOTA)

Benchmark	AUROC	Acc	F1
ABIDE / Autism	0.606	0.572	0.513
ABIDE / Age	0.877	0.804	0.805
ABIDE / Sex	0.694	0.736	0.834
ADNI / AD-HC	0.731	0.651	0.516
HCP / Sex	0.962	0.892	0.879
HCP / Age	0.680	0.618	0.597
COBRE / Schizophrenia	0.602	0.561	0.574

Only ADNI (Brain-JEPA) and ADHD-200 (NeuroSTORM) are same-dataset comparisons.

SOTA sources: Brain-JEPA (arXiv 2409.19407, Tables 2–3, fine-tuning; Acc/F1 only) · NeuroSTORM (arXiv 2506.11167).